

Project Scope Document

KijaniKiosk End-to-End Delivery Capstone

Problem Statement

KijaniKiosk previously lacked a production-grade delivery system: no environment separation, no reproducible infrastructure, no automated promotion flow, and no observability tied to service health. This made deployments manual, risky, and non-reproducible across environments.

Track Chosen

Track A: Infrastructure-First

Goal

Build a production-approaching delivery system where:

- Infrastructure is reproducible (Terraform + Ansible)
- Deployments are automated via CI/CD (Jenkins)
- Staging and production are isolated but consistent
- Observability signals detect service degradation
- Release promotion is gated by validation

In-Scope Components

- `kijani-staging` Kubernetes namespace provisioned via Terraform
- Configuration management via Ansible (namespace + config application)
- Single deployment manifest reused across staging and production
- Jenkins CI/CD pipeline:
 - auto deploy to staging on merge to `main`
 - smoke tests against staging
 - manual approval gate for production
- `kk-payments` deployed in both environments
- Environment-specific ConfigMaps (`DB_HOST` differs per env)
- Log-based error rate monitoring (SLO-style)

Out-of-Scope

- Full production Kubernetes HA tuning
- Service mesh implementation
- Multi-region deployments
- Advanced APM tooling (Datadog/New Relic)

Success Criteria

- One command (or pipeline run) can deploy system to staging
- Promotion to production is controlled and auditable
- A new engineer can deploy locally within 1 day using README
- Error rate can be observed from logs without external SaaS tools